

Q30

Let C be original number of counts due to the α -source.

Let B be the background count.

$$C + B = 300 \dots (1)$$

$$0.5C + B = 155 \dots (2)$$

Solving, $C = 290$ and $B = 10$.

Lead would block all the counts due to the α -source.

Hence, only background count of 10 is detected.

Ans: C

***** **END** *****